



AI CODING • STACK & CASE STUDY

AI coding stack + *WorldAI*

*Develop at **idea velocity** — a tour of the autonomous pipeline and the game I built with it.*

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SCAN TO CONNECT

From idea *to a merged PR*

- **Where else this works** Same stack, inside Snapchat.
- 01 **Origin story** AI Dungeon forgot my story. So I built one that doesn't.
- 02 **The story** A game session that went somewhere I never coded.
- 03 **The pipeline** Idea to merged PR — Hermes, one Slack message, a live session.
- 04 **How it ships** The agent pyramid, and where my attention actually goes.
- 05 **What you can take** The skills I use every day, open-sourced.
- 06 **WorldAI, in depth** The game, the rules underneath it, and the architecture.

How I got into *building with AI*

01

Started on AI Dungeon

It lost the plot by turn 50 — even on the \$50/month tier.

02

Built my own

Same failure in a week. The model was never the problem. Context was.

03

Survive 100 turns

Months of context budgeting. First to make it: a Star Wars AU where Luke joins Vader.

04

WorldAI

Past 1000 turns. Everything after this slide is how.

05

Then, the day job

Same harness, same habits — now at Snapchat.

Built AI world simulation → The player tried to quit. *The world said no.*

WorldAI is a live AI World simulator — the world reacts to plain language commands and remembers.

Playing with my mom, her character tried to walk away from the business

The world argued back — saying if she tried to run away she'd go to jail since she has a lease and debt.

She stayed, and twenty scenes later she turned around her business.

REAL TRANSCRIPT • MICHELE FRIED CHICKEN CAMPAIGN



WorldAI

Knowledge base / wiki

Reddit

Discord

Michele fried chicken

Spicy

"I'm done," you whisper, the words sounding foreign in the quiet shop. "I'm going home. It's too hard. It's just... too much."

Kim (Lvl 1) stares at you, her phone slipping from her fingers and hitting the tile with a dull thud. Her eyes, usually so sharp with aesthetic ambition, are wide with shock. "Michele? No. You can't. We just got the license! The regulars are outside right now fighting for us!" She takes a step toward you, her voice cracking. "If you leave now, Chloe wins. Julian wins. They'll say I really was part of a fraud in San Francisco. They'll say you were just another exploiter who ran when the heat got too high."

Marcus (Lvl 3) doesn't look up from the fryer he's scrubbing. His jaw is set so tight you can see the muscle jumping. "The debt to Mr. Park doesn't vanish because you close the door, Michele," he says, his voice a low, gravelly warning. "And the police have your name on that report. You try to 'go home' without settling the lease and the loan, and you're not just leaving a shop—you're becoming a fugitive from a paper trail you can't outrun."

One Slack message, *idea all the way to green PR* *(magic, when it works)*

A single natural-language message triggers the entire flow.
Bigger tasks still keep me in the loop.



110 contributions a day • 19 PRs merged a day • 6 agent CLIs • 23 models

No manual re-entry of context, no babysitting CI, no chasing reviewers. **Full automation is the north star; on bigger tasks I'm still in the middle.**

One AI *that talks to everything you do*

Three pieces — **frictionless I/O, central data, memory** — and the agent stack that runs underneath them.

P-01

Frictionless comms

“Talk to your AI the way you talk to a teammate.”

Triggered the same way you message a person. No special CLI, no IDE plugin.

P-02

Central data access

“The more access, the more powerful.”

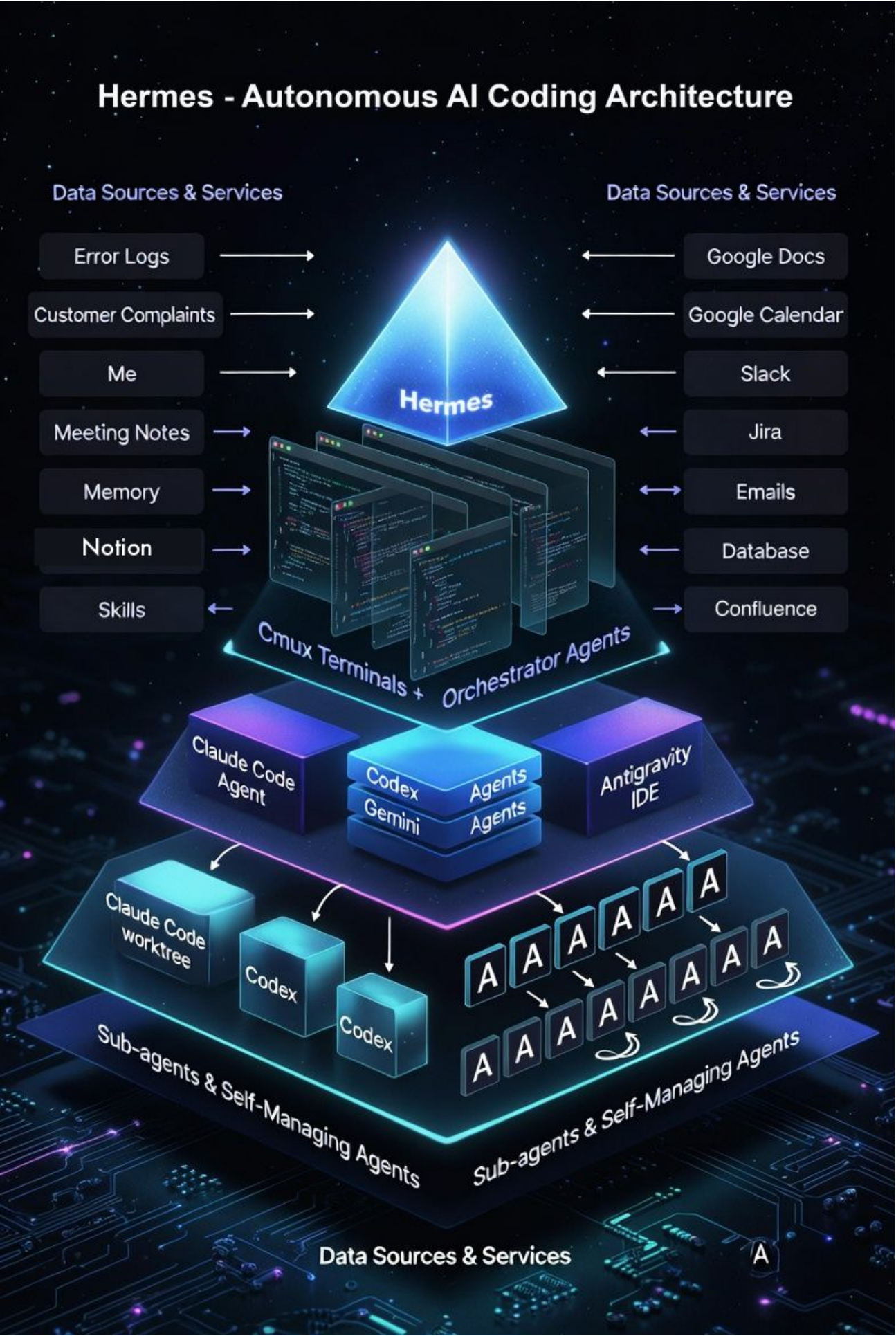
Hermes reads from everything you can read. Code is just one input.

P-03

Memory

“Replace more and more of your decisions over time.”

Each request reasons against your history, not a cold start.



"Let's test this technique *out*."

One thread → research → 99 replies of agent work.
Reply with the deliverable, no project tracking, no handoff.

01 · PROMPT

#WORLD AI · SLACK



Jeffrey Lee-Chan Today at 5:11 PM
Let's test this technique out. I want sprites for this campaign

<https://x.com/chongdashu/status/2054857868394832013?s=46>

Alexiel V2 ▶



Chong-U (@chongdashu) on X

I've been experimenting a lot with using AI for game sprites.
It's not perfect - but with the right pipeline you can get quite far.
Been building a suite of tools to enable you to achieve game ready sprites using
Images 2.0 & Nano Banana 2
Teaser of its capabilities below 📌
X (formerly Twitter) (39 kB) ▼



99 replies



hermes APP Today at 5:12 PM
Let me read the campaign doc and check out the technique from that X post.
Let me research the technique properly.
⌚ Still working... (3 min elapsed — iteration 5/60, running: delegate_task)

02 · RESULT

HERMES APP · 45 MIN LATER



ALEXIEL V3 — Tall/Lean Proportions
grok-image-image-quality | prompt: ...tall lean athletic build, NOT stocky...
45 minutes ago in # worldai - alexiel_v3_comparison.png



Hands on the keyboard, *Hermes on standby*

What you're seeing — *terminal multiplexer for agents*

The default is me — I sit in CMUX and drive.

Well-scoped work goes to Hermes: it sends keystrokes into a pane, watches the output, and reports back.

One pane isn't always one agent — sometimes it's a manager spawning sub-agents inside itself.

- manual by default
- hermes-driven on demand
- 1 pane → N agents
- per-pane git worktree
- evidence capture



The agent supervision pyramid — *top to bottom*

One cognitive task, tracked from dispatch down to tools.

01

Tier 01 · Apex — Dispatch (Hermes)

A Slack bot handles dispatch — routing, task queuing, goal coordination — backed by an HTTP API gateway. Retains long-term memory and preferences.

slack

memory

1 cognitive task

02

Tier 02 — Orchestration (CMUX terminal manager & Agent Orchestrator daemon)

Two background services manage terminal panes, process groups, isolated code checkouts (git worktrees), watchdog restarts, and automated pipelines.

terminal sessions

worktrees

multiple approaches

03

Tier 03 — **Workers** (CLI Agent Fleet)

Claude Code, OpenAI Codex, Cursor CLI. Specialized workers executing code changes in isolated worktree sandboxes.

claude-code

codex

long-running tasks

04

Tier 04 · Base — **Tools** (Subagents & MCP)

Short-lived tool specialists: Playwright browser testers, compiler drivers, BigQuery telemetry fetchers, test runners — some invoked via MCP (Model Context Protocol, a standard for AI-to-tool integration).

playwright

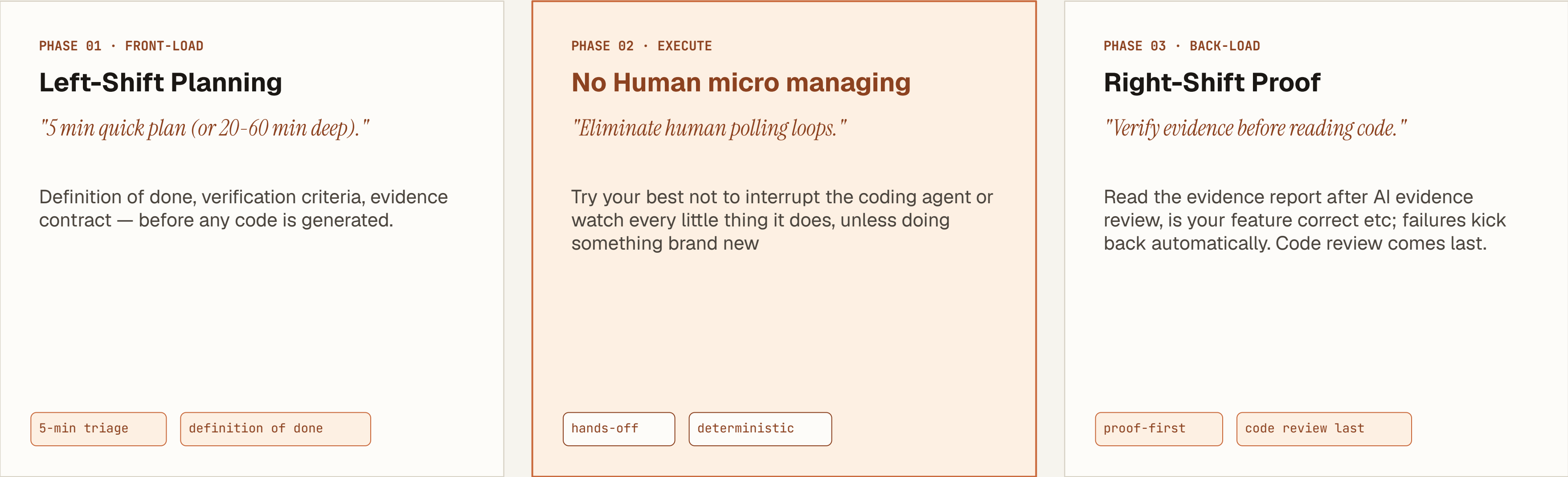
mcp

tool calls

Isolated checkouts: orchestrator supervises, workers execute, subagents test. **What matters isn't # agents — it's the # cognitive tasks + context switches**

Left-shift and right-shift, *no human micro managing in the middle*

Human focus allocation across the lifecycle. Front-load alignment, back-load verification.



Goal: **Cut human micro management to near-zero** — the human still owns the plan and the merge approval.

Shift left, shift right — *a portable skill system*

Six portable control points. WorldAI is what I built with them.

SHIFT LEFT • BEFORE THE CODE

/advice + /web-advice

Fan out independent reviewers before you commit to an approach.
/web-advice will use your browser window to avoid API cost

/repro + /root-cause-first

Recreate the exact failure on a real target, then fix its cause — not a symptom or a workaround.

/redgreen

Only a test you watch fail proves the bug is real.

SHIFT RIGHT • AFTER THE CODE

/4layer

unit → end-to-end → MCP/HTTP → browser. The first layer that reproduces the bug is the layer that owns it.

/evidence-review

Every claim resolves to a real artifact. Looking verified is not being verified.

/harness

When it goes wrong twice, fix the process that let it through.

Not shown: /web-advice, /dark-factory, /memory-search — real, but they depend on my own infrastructure (authenticated browsers, an external runner, a local memory store).

An AI Dungeon Master, playing live

Start a campaign in one sentence — the system generates the setting, your character, the rules, and the opening scene. The world reacts to what you do, and you don't always win.

D&D 5e

14 specialized agents

37+ prompt files

bring your own key

live at worldarchitect.ai

G • 01

D&D ruleset is essential

You don't always win. Roll random dice + probability range

G • 02

The world reacts and remembers

Factions act on their own goals. NPCs remember what you did weeks ago.

G • 03

God Mode — change anything

Rewrite your character, the world, or the rules in plain English. It persists.



worldarchitect.ai

WorldAI

Knowledge base / wiki

Reddit

Discord

✕

Dragon Knight

ⓘ

Spicy

Back to Dashboard

Timestamp: 95 A.G., Frost-Fall 12, 09:00:00
Location: The King's Ribbon, Winter-Mourn Province
Status: Lvl 1 Paladin | HP: 12/12 | Gold: 25gp | XP: 0/300
Conditions: Normal | Exhaustion: 0 | Inspiration: No

Equipment

Stats

Spells

Resources:

HD: 1/1, Lay on Hands: 5/5, Divine Sense: 4/4

Dice Rolls:

• 1d20+1 = 18 vs DC 12 - Success (Perception check to scout Winter-Mourn Keep and the refugee settlement)

REWARDS (encounter):

+10 XP | XP: 10/300 (0%)

Scene #3:

Gisling your horse forward, you leave the steady rhythm of the main detachment behind, the white stone of the King's Ribbon blurring as you ascend a jagged limestone ridge overlooking the valley. From this vantage point, Winter-Mourn Keep looms like a tooth of gray granite. Your eyes, trained by the marshals of House Valerion, scan the defenses. The primary gatehouse is manned by four sentries in mismatched leather—local militia, not Imperial regulars. They look tired, their attention focused more on the cookfires than the road.

Below the walls, the refugee settlement is a sprawling misery of canvas and timber. It is no military encampment. You see elderly women mending nets, children chasing a mangy dog between the shacks, and a line of people waiting for a meager ration from a central cauldron. There are no barricades here, no hidden caches of pikes—only the desperate clinging to the shadow of the keep. To the western side of the fortress, you spot a narrow postern gate, partially obscured by a frozen waterfall. It is unattended, a tactical lapse that suggests Lady Annalise Ashwood's forces are stretched thin or perhaps overly reliant on the province's historical peace.

Time to first token: 10.9s, server gen: 9.5s, total: 20.8s

4 options — expand to see what you can do

Tap to expand

Custom Action: describe whatever you want and your character will do it...

Character ⓘ

Think/Plan ⓘ

God ⓘ

Send

Real session at worldarchitect.ai — server-resolved dice roll, XP award, and generated scene.

Query-aware routing, token budget, *implicit caching*.

T•01

Context management

Up to 300k tokens, bring your own key. A Min-first / Fill-to-max budget allocator: 40% story · 20% system · 5% memories · 3% npcs, remainder spills into story.

300k tokens · 4 components · bring your own key

T•02

Two-layer inference

Layer 1: FastEmbed, a small embedding model on my own machine, routes the request to an agent by meaning — local and free. Layer 2: Gemini 3 Flash, so only the right agent's prompt hits the paid model.

<50ms local L1 · 14 agents · Gemini 3 Flash

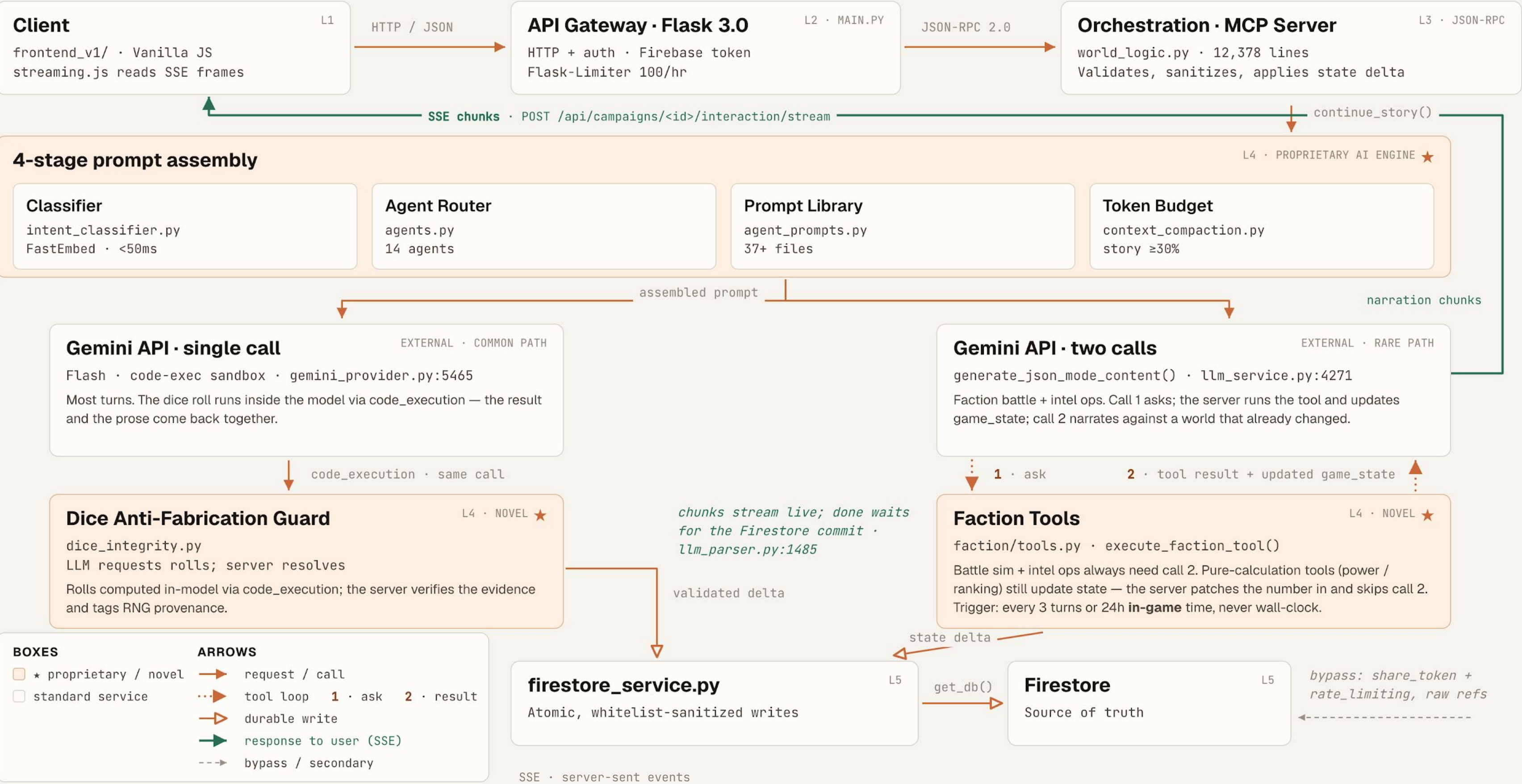
T•03

Implicit caching

A 537,444-char prompt prefix held byte-identical through turn 11 — checked at turns 1, 5, 9, 10 and 11. Cache hits are automatic when the prefix doesn't move; the engineering is keeping it still.

537k prefix · stable to turn 11 · 59% spend cut

Two engines, *one gateway*



What I built with this stack — *inside Snap*

Before AI: Snap Growth Notifications, scaled to 5B+ notifications a day, ~100x growth from 2018 to 2026.

Before Snap: YouTube Partner Program (2008–2018) — founding architect. **\$70B+ paid to creators, 2M+ partners onboarded, \$10B+ annual revenue at peak.**

Snap Bridge

MCP (Model Context Protocol) connector to internal services

BEFORE

Engineers copy-paste between a browser tab and an AI chat window to pull internal data.

AFTER

Stay in the terminal. The AI acts autonomously against internal services on your behalf and can drive tasks end2end.

Adopted by the majority of engineers

Snap Wings

Remote-hosting for internal apps

BEFORE

No path to remote-host an internal app — you either found a server or you didn't ship.

AFTER

A shared server auto-registers and deploys your app — like a one-click internal app-hosting service.

Used by engineers and non-engineers

Open Model Pilot WIP

Open-weight models as a frontier substitute

STATUS

Internal benchmarks currently just matching SWE bench/online results, nothing interesting yet

GAP

Most open-model providers still lack full Claude Code / Codex API-level tool-use compatibility.

Directional, not yet a verdict

Play the game, find me

★ PLAY

WorldAI *D&D living world simulation*

Start a campaign in one sentence. Free for now — bring your own key.



worldarchitect.ai

01 · CONNECT

LinkedIn

Let me know if you have any feedback on my game or harness!

linkedin.com/in/jeffrey-lee-chan

02 · BUILD

dark-factory (WIP) *one command → green PR*

One slash command drives a task from a spec to a green PR.

github.com/jleechanorg/dark-factory

03 · CODE

Agent Skills *left and right shift your life!*

My open-source agentic skills, including the harness that built this deck.

github.com/jleechanorg/jleechan-skills

Thanks for watching. Now go ship something at idea velocity.

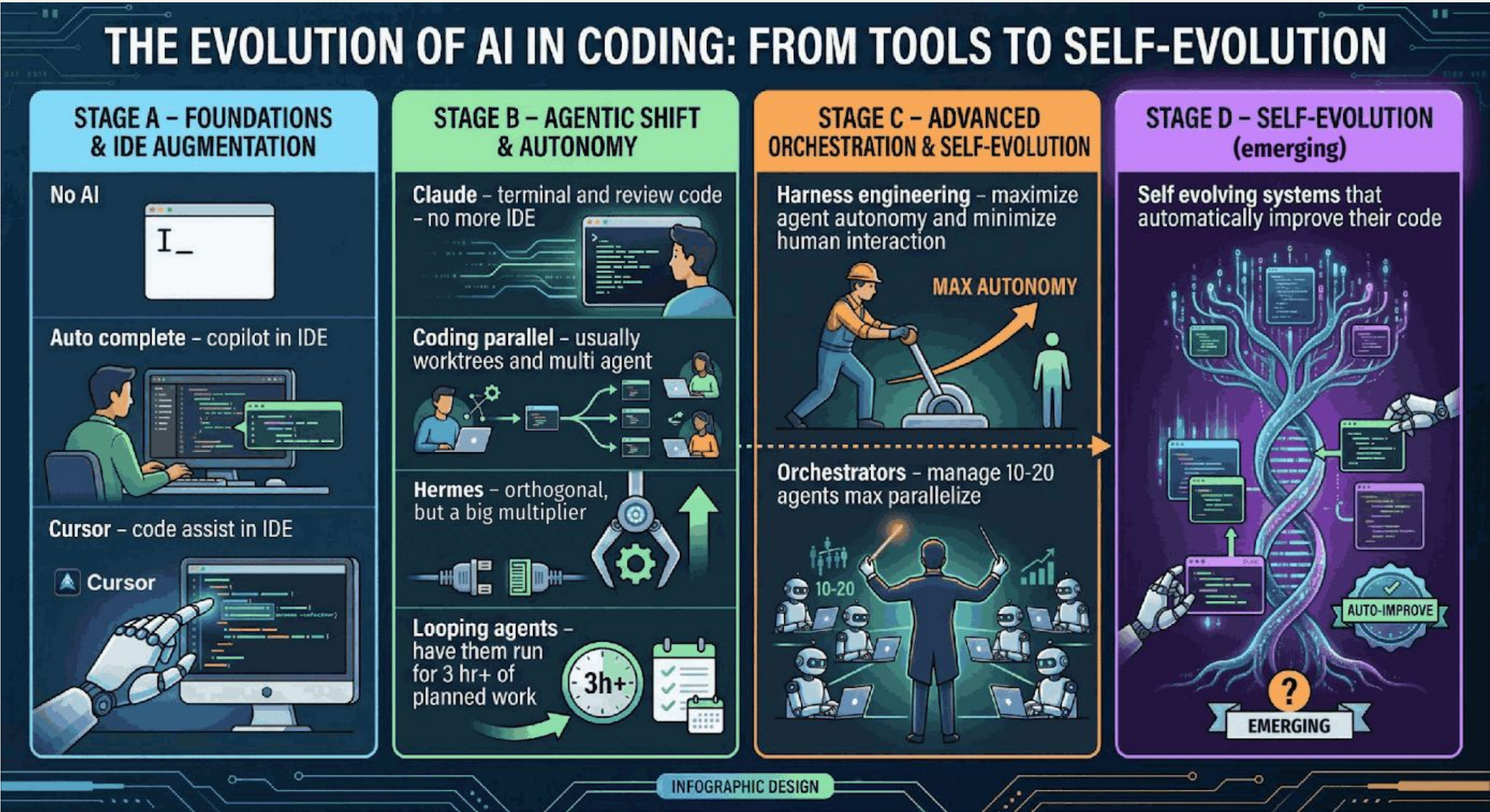
§ APPENDIX

Appendix — *the loop, and how AI coding got here*

From tools *to self-evolution*

Four stages, less human at each.
Most teams I've worked with sit at B. I'm running C.

STAGE A → B → C → D



The 6-step loop — *idea to verified merge*

